

a Client updates $u_1, \dots, u_n$ of round r

Stage 1: geometric-median acceptance region

m = geometric median of all updates (Weiszfeld)
 $B_0 = \{i: \|u_i - m\| \leq 2.5 \times \text{median distance}\}$
 reset $B_0 = U$ if $|B_0| < n/2$

Stage 2: relative-density gate

$rd_i = k' / \sum$ of k' -NN distances, $k' = 5$
 opens if a density gap (max/median > 10) coincides
 with $\geq \max(2, \lceil 0.05n \rceil)$ Stage-1 rejections; 3-round memory

gate open

Stage 3: cluster validation

SNN clustering of the low-density set $L = \{i: rd_i < t\}$
 reject group S if $\|c_S - m_{B_0}\| > \tau R_0$, $\tau = 2$
 R_0 = median distance to m_{B_0} within B_0

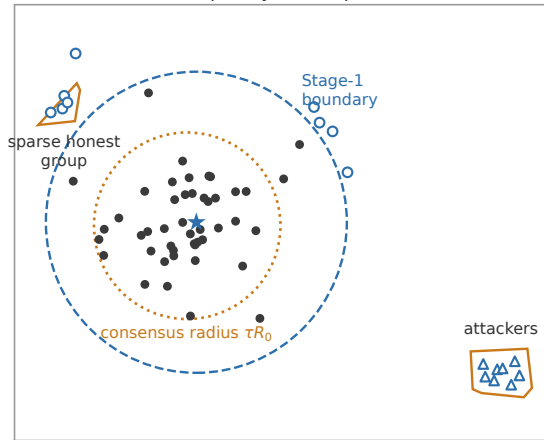
Combine: $B = B_0 \setminus (\text{rejected groups})$

valve: if $|B| < n/2$, restore $B = B_0$

aggregate: $w_{r+1} = w_r + \text{mean of } u_i \text{ over } i \in B$

b

Illustrative 2-D example (synthetic points)



- honest, accepted
- honest, excluded
- ★ geometric median m
- △ attacker, excluded
- group rejected by Stage 3

gate closed:
 $B = B_0$